In [1]:
```
# ============================================================
# MEMBER 3 — EDA & STATISTICAL ANALYSIS
# STEPS 22–56
# ============================================================

# Prerequisite: Run Member 2 first so the cleaned CSV exists.
# This script independently loads the cleaned CSV from the expected output path.
# It also recreates the common output folders if needed.

import os
import warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import joblib
warnings.filterwarnings('ignore')
sns.set_theme(style='whitegrid')

INPUT_FILE = (r"C:\Users\admin\OneDrive\Desktop\Hotel Booking Cancellation Predi
OUTPUT_FOLDER = 'output'
PREPROCESSING_FOLDER = os.path.join(OUTPUT_FOLDER, 'preprocessing_results')
EDA_FOLDER = os.path.join(OUTPUT_FOLDER, 'eda_results')
VISUALIZATION_FOLDER = os.path.join(OUTPUT_FOLDER, 'visualization_results')
ML_FOLDER = os.path.join(OUTPUT_FOLDER, 'machine_learning_results')
for folder in [OUTPUT_FOLDER, PREPROCESSING_FOLDER, EDA_FOLDER, VISUALIZATION_FO
    os.makedirs(folder, exist_ok=True)

CLEANED_FILE = os.path.join(PREPROCESSING_FOLDER, 'hotel_bookings_cleaned.csv')
if not os.path.exists(CLEANED_FILE):
    raise FileNotFoundError('Run Member 2 first; cleaned dataset not found: ' +
df = pd.read_csv(CLEANED_FILE)
month_map = {'January': 1, 'February': 2, 'March': 3, 'April': 4, 'May': 5, 'Jun
target_column = 'is_canceled'
numeric_columns = df.select_dtypes(include=np.number).columns.tolist()
categorical_columns = df.select_dtypes(exclude=np.number).columns.tolist()

# ============================================================
# STEP 23 — BASIC DATASET INFORMATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 23 — BASIC DATASET INFORMATION")
print("=" * 70)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

print("\nDataset Information:")
df.info()
```

```
======================================================================
STEP 23 — BASIC DATASET INFORMATION
======================================================================

Column Names:
['hotel', 'is_canceled', 'lead_time', 'arrival_date_year', 'arrival_date_month',
'arrival_date_week_number', 'arrival_date_day_of_month', 'stays_in_weekend_night
s', 'stays_in_week_nights', 'adults', 'children', 'babies', 'meal', 'country', 'm
arket_segment', 'distribution_channel', 'is_repeated_guest', 'previous_cancellati
ons', 'previous_bookings_not_canceled', 'reserved_room_type', 'assigned_room_typ
e', 'booking_changes', 'deposit_type', 'agent', 'company', 'days_in_waiting_lis
t', 'customer_type', 'adr', 'required_car_parking_spaces', 'total_of_special_requ
ests', 'reservation_status', 'reservation_status_date', 'city', 'total_nights',
'total_guests', 'arrival_month_num', 'arrival_date']

Data Types:
hotel                             str
is_canceled                     int64
lead_time                       int64
arrival_date_year               int64
arrival_date_month                str
arrival_date_week_number        int64
arrival_date_day_of_month       int64
stays_in_weekend_nights         int64
stays_in_week_nights            int64
adults                          int64
children                      float64
babies                          int64
meal                              str
country                           str
market_segment                    str
distribution_channel              str
is_repeated_guest               int64
previous_cancellations          int64
previous_bookings_not_canceled  int64
reserved_room_type                str
assigned_room_type                str
booking_changes                 int64
deposit_type                      str
agent                         float64
company                       float64
days_in_waiting_list            int64
customer_type                     str
adr                           float64
required_car_parking_spaces     int64
total_of_special_requests       int64
reservation_status                str
reservation_status_date           str
city                              str
total_nights                    int64
total_guests                  float64
arrival_month_num               int64
arrival_date                      str
dtype: object

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 37 columns):
 #   Column                          Non-Null Count   Dtype
```

```
---  ------                          --------------   -----
 0   hotel                           119390 non-null  str
 1   is_canceled                     119390 non-null  int64
 2   lead_time                       119390 non-null  int64
 3   arrival_date_year               119390 non-null  int64
 4   arrival_date_month              119390 non-null  str
 5   arrival_date_week_number        119390 non-null  int64
 6   arrival_date_day_of_month       119390 non-null  int64
 7   stays_in_weekend_nights         119390 non-null  int64
 8   stays_in_week_nights            119390 non-null  int64
 9   adults                          119390 non-null  int64
 10  children                        119390 non-null  float64
 11  babies                          119390 non-null  int64
 12  meal                            119390 non-null  str
 13  country                         119390 non-null  str
 14  market_segment                  119390 non-null  str
 15  distribution_channel            119390 non-null  str
 16  is_repeated_guest               119390 non-null  int64
 17  previous_cancellations          119390 non-null  int64
 18  previous_bookings_not_canceled  119390 non-null  int64
 19  reserved_room_type              119390 non-null  str
 20  assigned_room_type              119390 non-null  str
 21  booking_changes                 119390 non-null  int64
 22  deposit_type                    119390 non-null  str
 23  agent                           119390 non-null  float64
 24  company                         119390 non-null  float64
 25  days_in_waiting_list            119390 non-null  int64
 26  customer_type                   119390 non-null  str
 27  adr                             119390 non-null  float64
 28  required_car_parking_spaces     119390 non-null  int64
 29  total_of_special_requests       119390 non-null  int64
 30  reservation_status              119390 non-null  str
 31  reservation_status_date         119390 non-null  str
 32  city                            119390 non-null  str
 33  total_nights                    119390 non-null  int64
 34  total_guests                    119390 non-null  float64
 35  arrival_month_num               119390 non-null  int64
 36  arrival_date                    119390 non-null  str
dtypes: float64(5), int64(18), str(14)
memory usage: 33.7 MB
```

In [2]:

```
# ============================================================
# STEP 24 — MISSING VALUE ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 24 — MISSING VALUE ANALYSIS")
print("=" * 70)

missing_count = df.isnull().sum()
missing_percentage = (missing_count / len(df)) * 100

missing_analysis = pd.DataFrame({
    "Missing_Count": missing_count,
    "Missing_Percentage": missing_percentage
})

print(missing_analysis)

missing_analysis.to_csv(
```

```
    os.path.join(
        EDA_FOLDER,
        "step_24_missing_value_analysis.csv"
    )
)
```

```
======================================================================
STEP 24 — MISSING VALUE ANALYSIS
======================================================================
                                Missing_Count  Missing_Percentage
hotel                                       0                 0.0
is_canceled                                 0                 0.0
lead_time                                   0                 0.0
arrival_date_year                           0                 0.0
arrival_date_month                          0                 0.0
arrival_date_week_number                    0                 0.0
arrival_date_day_of_month                   0                 0.0
stays_in_weekend_nights                     0                 0.0
stays_in_week_nights                        0                 0.0
adults                                      0                 0.0
children                                    0                 0.0
babies                                      0                 0.0
meal                                        0                 0.0
country                                     0                 0.0
market_segment                              0                 0.0
distribution_channel                        0                 0.0
is_repeated_guest                           0                 0.0
previous_cancellations                      0                 0.0
previous_bookings_not_canceled              0                 0.0
reserved_room_type                          0                 0.0
assigned_room_type                          0                 0.0
booking_changes                             0                 0.0
deposit_type                                0                 0.0
agent                                       0                 0.0
company                                     0                 0.0
days_in_waiting_list                        0                 0.0
customer_type                               0                 0.0
adr                                         0                 0.0
required_car_parking_spaces                 0                 0.0
total_of_special_requests                   0                 0.0
reservation_status                          0                 0.0
reservation_status_date                     0                 0.0
city                                        0                 0.0
total_nights                                0                 0.0
total_guests                                0                 0.0
arrival_month_num                           0                 0.0
arrival_date                                0                 0.0
```

In [3]:
```
# ============================================================
# STEP 25 — DUPLICATE ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 25 — DUPLICATE ANALYSIS")
print("=" * 70)

print("Duplicate records:", df.duplicated().sum())
```

```
======================================================================
STEP 25 — DUPLICATE ANALYSIS
======================================================================
Duplicate records: 0
```

In [4]:
```
# ============================================================
# STEP 26 — IDENTIFY NUMERICAL FEATURES
# ============================================================

print("\n" + "=" * 70)
print("STEP 26 — NUMERICAL FEATURES")
print("=" * 70)

numeric_columns = df.select_dtypes(
    include=np.number
).columns.tolist()

print("Numerical columns:")
print(numeric_columns)
```

```
======================================================================
STEP 26 — NUMERICAL FEATURES
======================================================================
Numerical columns:
['is_canceled', 'lead_time', 'arrival_date_year', 'arrival_date_week_number', 'ar
rival_date_day_of_month', 'stays_in_weekend_nights', 'stays_in_week_nights', 'adu
lts', 'children', 'babies', 'is_repeated_guest', 'previous_cancellations', 'previ
ous_bookings_not_canceled', 'booking_changes', 'agent', 'company', 'days_in_waiti
ng_list', 'adr', 'required_car_parking_spaces', 'total_of_special_requests', 'tot
al_nights', 'total_guests', 'arrival_month_num']
```

In [5]:
```
# ============================================================
# STEP 27 — IDENTIFY CATEGORICAL FEATURES
# ============================================================

print("\n" + "=" * 70)
print("STEP 27 — CATEGORICAL FEATURES")
print("=" * 70)

categorical_columns = df.select_dtypes(
    exclude=np.number
).columns.tolist()

print("Categorical columns:")
print(categorical_columns)
```

```
======================================================================
STEP 27 — CATEGORICAL FEATURES
======================================================================
Categorical columns:
['hotel', 'arrival_date_month', 'meal', 'country', 'market_segment', 'distributio
n_channel', 'reserved_room_type', 'assigned_room_type', 'deposit_type', 'customer
_type', 'reservation_status', 'reservation_status_date', 'city', 'arrival_date']
```

In [6]:
```
# ============================================================
# STEP 28 — DESCRIPTIVE STATISTICS
# ============================================================

print("\n" + "=" * 70)
print("STEP 28 — DESCRIPTIVE STATISTICS")
print("=" * 70)
```

```
descriptive_statistics = df.describe().T

print(descriptive_statistics)

descriptive_statistics.to_csv(
    os.path.join(
        EDA_FOLDER,
        "step_28_descriptive_statistics.csv"
    )
)
```

```
======================================================================
STEP 28 — DESCRIPTIVE STATISTICS
======================================================================
                                  count         mean         std      min  \
is_canceled                    119390.0     0.370416    0.482918     0.00
lead_time                      119390.0   104.011416  106.863097     0.00
arrival_date_year              119390.0  2024.000000    0.000000  2024.00
arrival_date_week_number       119390.0    26.375835   15.021596     1.00
arrival_date_day_of_month      119390.0    15.723394    8.805079     1.00
stays_in_weekend_nights        119390.0     0.927599    0.998613     0.00
stays_in_week_nights           119390.0     2.500302    1.908286     0.00
adults                         119390.0     1.856403    0.579261     0.00
children                       119390.0     0.103886    0.398555     0.00
babies                         119390.0     0.007949    0.097436     0.00
is_repeated_guest              119390.0     0.031912    0.175767     0.00
previous_cancellations         119390.0     0.087118    0.844336     0.00
previous_bookings_not_canceled 119390.0     0.137097    1.497437     0.00
booking_changes                119390.0     0.221124    0.652306     0.00
agent                          119390.0    74.828319  107.141953     0.00
company                        119390.0    10.775157   53.943884     0.00
days_in_waiting_list           119390.0     2.321149   17.594721     0.00
adr                            119390.0   101.831122   50.535790    -6.38
required_car_parking_spaces    119390.0     0.062518    0.245291     0.00
total_of_special_requests      119390.0     0.571363    0.792798     0.00
total_nights                   119390.0     3.427900    2.557439     0.00
total_guests                   119390.0     1.968239    0.722394     0.00
arrival_month_num              119390.0     6.484136    3.437030     1.00

                                   25%       50%     75%     max
is_canceled                       0.00     0.000     1.0     1.0
lead_time                        18.00    69.000   160.0   737.0
arrival_date_year              2024.00  2024.000  2024.0  2024.0
arrival_date_week_number         13.00    26.000    39.0    52.0
arrival_date_day_of_month         8.00    16.000    23.0    31.0
stays_in_weekend_nights           0.00     1.000     2.0    19.0
stays_in_week_nights              1.00     2.000     3.0    50.0
adults                            2.00     2.000     2.0    55.0
children                          0.00     0.000     0.0    10.0
babies                            0.00     0.000     0.0    10.0
is_repeated_guest                 0.00     0.000     0.0     1.0
previous_cancellations            0.00     0.000     0.0    26.0
previous_bookings_not_canceled    0.00     0.000     0.0    72.0
booking_changes                   0.00     0.000     0.0    21.0
agent                             7.00     9.000   152.0   535.0
company                           0.00     0.000     0.0   543.0
days_in_waiting_list              0.00     0.000     0.0   391.0
adr                              69.29    94.575   126.0  5400.0
required_car_parking_spaces       0.00     0.000     0.0     8.0
total_of_special_requests         0.00     0.000     1.0     5.0
total_nights                      2.00     3.000     4.0    69.0
total_guests                      2.00     2.000     2.0    55.0
arrival_month_num                 3.00     7.000     9.0    12.0
```

In [7]:
```python
# ============================================================
# STEP 29 — MEAN CALCULATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 29 — MEAN")
print("=" * 70)
```

```
mean_values = df[numeric_columns].mean()

print(mean_values)

mean_values.to_csv(
    os.path.join(EDA_FOLDER, "step_29_mean.csv"),
    header=["Mean"]
)
```

```
======================================================================
STEP 29 — MEAN
======================================================================
is_canceled                          0.370416
lead_time                          104.011416
arrival_date_year                 2024.000000
arrival_date_week_number            26.375835
arrival_date_day_of_month           15.723394
stays_in_weekend_nights              0.927599
stays_in_week_nights                 2.500302
adults                               1.856403
children                             0.103886
babies                               0.007949
is_repeated_guest                    0.031912
previous_cancellations               0.087118
previous_bookings_not_canceled       0.137097
booking_changes                      0.221124
agent                               74.828319
company                             10.775157
days_in_waiting_list                 2.321149
adr                                101.831122
required_car_parking_spaces          0.062518
total_of_special_requests            0.571363
total_nights                         3.427900
total_guests                         1.968239
arrival_month_num                    6.484136
dtype: float64
```

In [8]:
```
# ============================================================
# STEP 30 — MEDIAN CALCULATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 30 — MEDIAN")
print("=" * 70)

median_values = df[numeric_columns].median()

print(median_values)

median_values.to_csv(
    os.path.join(EDA_FOLDER, "step_30_median.csv"),
    header=["Median"]
)
```

```
======================================================================
STEP 30 — MEDIAN
======================================================================
is_canceled                          0.000
lead_time                           69.000
arrival_date_year                 2024.000
arrival_date_week_number            26.000
arrival_date_day_of_month           16.000
stays_in_weekend_nights              1.000
stays_in_week_nights                 2.000
adults                               2.000
children                             0.000
babies                               0.000
is_repeated_guest                    0.000
previous_cancellations               0.000
previous_bookings_not_canceled       0.000
booking_changes                      0.000
agent                                9.000
company                              0.000
days_in_waiting_list                 0.000
adr                                 94.575
required_car_parking_spaces          0.000
total_of_special_requests            0.000
total_nights                         3.000
total_guests                         2.000
arrival_month_num                    7.000
dtype: float64
```

In [9]:

```
# ============================================================
# STEP 31 — MODE CALCULATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 31 — MODE")
print("=" * 70)

mode_values = df[numeric_columns].mode()

print(mode_values)

mode_values.to_csv(
    os.path.join(EDA_FOLDER, "step_31_mode.csv"),
    index=False
)
```

```
======================================================================
STEP 31 — MODE
======================================================================
   is_canceled  lead_time  arrival_date_year  arrival_date_week_number  \
0            0          0               2024                         1

   arrival_date_day_of_month  stays_in_weekend_nights  stays_in_week_nights  \
0                         15                       0                     2

   adults  children  babies  ...  booking_changes  agent  company  \
0       2       0.0       0  ...                0    9.0      0.0

   days_in_waiting_list   adr  required_car_parking_spaces  \
0                     0  62.0                            0

   total_of_special_requests  total_nights  total_guests  arrival_month_num
0                          0             2           2.0                 10

[1 rows x 23 columns]
```

In [10]:

```
# ============================================================
# STEP 32 — STANDARD DEVIATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 32 — STANDARD DEVIATION")
print("=" * 70)

std_values = df[numeric_columns].std()

print(std_values)

std_values.to_csv(
    os.path.join(EDA_FOLDER, "step_32_standard_deviation.csv"),
    header=["Standard_Deviation"]
)
```

```
======================================================================
STEP 32 — STANDARD DEVIATION
======================================================================
is_canceled                          0.482918
lead_time                          106.863097
arrival_date_year                    0.000000
arrival_date_week_number            15.021596
arrival_date_day_of_month            8.805079
stays_in_weekend_nights              0.998613
stays_in_week_nights                 1.908286
adults                               0.579261
children                             0.398555
babies                               0.097436
is_repeated_guest                    0.175767
previous_cancellations               0.844336
previous_bookings_not_canceled       1.497437
booking_changes                      0.652306
agent                              107.141953
company                             53.943884
days_in_waiting_list                17.594721
adr                                 50.535790
required_car_parking_spaces          0.245291
total_of_special_requests            0.792798
total_nights                         2.557439
total_guests                         0.722394
arrival_month_num                    3.437030
dtype: float64
```

In [11]:
```
# ============================================================
# STEP 33 — VARIANCE
# ============================================================

print("\n" + "=" * 70)
print("STEP 33 — VARIANCE")
print("=" * 70)

variance_values = df[numeric_columns].var()

print(variance_values)

variance_values.to_csv(
    os.path.join(EDA_FOLDER, "step_33_variance.csv"),
    header=["Variance"]
)
```

```
======================================================================
STEP 33 — VARIANCE
======================================================================
is_canceled                          0.233210
lead_time                        11419.721511
arrival_date_year                    0.000000
arrival_date_week_number           225.648350
arrival_date_day_of_month           77.529415
stays_in_weekend_nights              0.997229
stays_in_week_nights                 3.641554
adults                               0.335543
children                             0.158846
babies                               0.009494
is_repeated_guest                    0.030894
previous_cancellations               0.712904
previous_bookings_not_canceled       2.242317
booking_changes                      0.425503
agent                            11479.398119
company                           2909.942594
days_in_waiting_list               309.574203
adr                               2553.866100
required_car_parking_spaces          0.060168
total_of_special_requests            0.628529
total_nights                         6.540493
total_guests                         0.521853
arrival_month_num                   11.813178
dtype: float64
```

In [12]:
```python
# ============================================================
# STEP 34 — MINIMUM VALUES
# ============================================================

print("\n" + "=" * 70)
print("STEP 34 — MINIMUM VALUES")
print("=" * 70)

minimum_values = df[numeric_columns].min()

print(minimum_values)

minimum_values.to_csv(
    os.path.join(EDA_FOLDER, "step_34_minimum.csv"),
    header=["Minimum"]
)
```

```
======================================================================
STEP 34 — MINIMUM VALUES
======================================================================
is_canceled                          0.00
lead_time                            0.00
arrival_date_year                 2024.00
arrival_date_week_number             1.00
arrival_date_day_of_month            1.00
stays_in_weekend_nights              0.00
stays_in_week_nights                 0.00
adults                               0.00
children                             0.00
babies                               0.00
is_repeated_guest                    0.00
previous_cancellations               0.00
previous_bookings_not_canceled       0.00
booking_changes                      0.00
agent                                0.00
company                              0.00
days_in_waiting_list                 0.00
adr                                 -6.38
required_car_parking_spaces          0.00
total_of_special_requests            0.00
total_nights                         0.00
total_guests                         0.00
arrival_month_num                    1.00
dtype: float64
```

In [13]:

```python
# ============================================================
# STEP 35 — MAXIMUM VALUES
# ============================================================

print("\n" + "=" * 70)
print("STEP 35 — MAXIMUM VALUES")
print("=" * 70)

maximum_values = df[numeric_columns].max()

print(maximum_values)

maximum_values.to_csv(
    os.path.join(EDA_FOLDER, "step_35_maximum.csv"),
    header=["Maximum"]
)
```

```
======================================================================
STEP 35 — MAXIMUM VALUES
======================================================================
is_canceled                          1.0
lead_time                          737.0
arrival_date_year                 2024.0
arrival_date_week_number            52.0
arrival_date_day_of_month           31.0
stays_in_weekend_nights             19.0
stays_in_week_nights                50.0
adults                              55.0
children                            10.0
babies                              10.0
is_repeated_guest                    1.0
previous_cancellations              26.0
previous_bookings_not_canceled      72.0
booking_changes                     21.0
agent                              535.0
company                            543.0
days_in_waiting_list               391.0
adr                               5400.0
required_car_parking_spaces          8.0
total_of_special_requests            5.0
total_nights                        69.0
total_guests                        55.0
arrival_month_num                   12.0
dtype: float64
```

In [14]:
```
# ============================================================
# STEP 36 — RANGE CALCULATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 36 — RANGE")
print("=" * 70)

range_values = (
    df[numeric_columns].max()
    - df[numeric_columns].min()
)

print(range_values)

range_values.to_csv(
    os.path.join(EDA_FOLDER, "step_36_range.csv"),
    header=["Range"]
)
```

```
======================================================================
STEP 36 — RANGE
======================================================================
is_canceled                          1.00
lead_time                          737.00
arrival_date_year                    0.00
arrival_date_week_number            51.00
arrival_date_day_of_month           30.00
stays_in_weekend_nights             19.00
stays_in_week_nights                50.00
adults                              55.00
children                            10.00
babies                              10.00
is_repeated_guest                    1.00
previous_cancellations              26.00
previous_bookings_not_canceled      72.00
booking_changes                     21.00
agent                              535.00
company                            543.00
days_in_waiting_list               391.00
adr                               5406.38
required_car_parking_spaces          8.00
total_of_special_requests            5.00
total_nights                        69.00
total_guests                        55.00
arrival_month_num                   11.00
dtype: float64
```

In [15]:
```python
# ============================================================
# STEP 37 — QUARTILE ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 37 — QUARTILES")
print("=" * 70)

quartiles = df[numeric_columns].quantile([0.25, 0.50, 0.75])

print(quartiles)

quartiles.to_csv(
    os.path.join(EDA_FOLDER, "step_37_quartiles.csv")
)
```

```
======================================================================
STEP 37 — QUARTILES
======================================================================
      is_canceled  lead_time  arrival_date_year  arrival_date_week_number  \
0.25          0.0       18.0             2024.0                      13.0
0.50          0.0       69.0             2024.0                      26.0
0.75          1.0      160.0             2024.0                      39.0

      arrival_date_day_of_month  stays_in_weekend_nights  \
0.25                        8.0                     0.0
0.50                       16.0                     1.0
0.75                       23.0                     2.0

      stays_in_week_nights  adults  children  babies  ...  booking_changes  \
0.25                   1.0     2.0       0.0     0.0  ...              0.0
0.50                   2.0     2.0       0.0     0.0  ...              0.0
0.75                   3.0     2.0       0.0     0.0  ...              0.0

      agent  company  days_in_waiting_list      adr  \
0.25    7.0      0.0                   0.0   69.290
0.50    9.0      0.0                   0.0   94.575
0.75  152.0      0.0                   0.0  126.000

      required_car_parking_spaces  total_of_special_requests  total_nights  \
0.25                          0.0                        0.0           2.0
0.50                          0.0                        0.0           3.0
0.75                          0.0                        1.0           4.0

      total_guests  arrival_month_num
0.25           2.0                3.0
0.50           2.0                7.0
0.75           2.0                9.0

[3 rows x 23 columns]
```

In [16]:

```python
# ============================================================
# STEP 38 — SKEWNESS ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 38 — SKEWNESS")
print("=" * 70)

skewness = df[numeric_columns].skew()

print(skewness.sort_values(ascending=False))

skewness.to_csv(
    os.path.join(EDA_FOLDER, "step_38_skewness.csv"),
    header=["Skewness"]
)
```

```
======================================================================
STEP 38 — SKEWNESS
======================================================================
babies                              24.646545
previous_cancellations              24.458049
previous_bookings_not_canceled      23.539800
adults                              18.317805
days_in_waiting_list                11.944353
adr                                 10.530214
total_guests                        10.154167
booking_changes                      6.000270
company                              5.918406
is_repeated_guest                    5.326315
required_car_parking_spaces          4.163233
children                             4.112670
total_nights                         3.308771
stays_in_week_nights                 2.862249
stays_in_weekend_nights              1.380046
total_of_special_requests            1.349189
lead_time                            1.346550
agent                                1.300287
is_canceled                          0.536678
arrival_date_week_number             0.003881
arrival_date_day_of_month            0.002373
arrival_date_year                    0.000000
arrival_month_num                   -0.001028
dtype: float64
```

In [17]:

```python
# ============================================================
# STEP 39 — KURTOSIS ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 39 — KURTOSIS")
print("=" * 70)

kurtosis = df[numeric_columns].kurtosis()

print(kurtosis.sort_values(ascending=False))

kurtosis.to_csv(
    os.path.join(EDA_FOLDER, "step_39_kurtosis.csv"),
    header=["Kurtosis"]
)
```

```
======================================================================
STEP 39 — KURTOSIS
======================================================================
babies                              1633.948235
adults                              1352.115116
adr                                 1013.189851
previous_bookings_not_canceled       767.245210
previous_cancellations               674.073693
total_guests                         554.650474
days_in_waiting_list                 186.793070
booking_changes                       79.393605
company                               37.753758
required_car_parking_spaces           29.998056
total_nights                          28.873721
is_repeated_guest                     26.370077
stays_in_week_nights                  24.284555
children                              18.674464
stays_in_weekend_nights                7.174066
lead_time                              1.696449
total_of_special_requests              1.492565
agent                                  0.512689
arrival_date_year                      0.000000
arrival_date_day_of_month             -1.200683
arrival_date_week_number              -1.200734
arrival_month_num                     -1.206790
is_canceled                           -1.712005
dtype: float64
```

In [18]:

```
# ============================================================
# STEP 40 — DISTRIBUTION ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 40 — DISTRIBUTION ANALYSIS")
print("=" * 70)

distribution_summary = pd.DataFrame({
    "Mean": df[numeric_columns].mean(),
    "Median": df[numeric_columns].median(),
    "Std_Dev": df[numeric_columns].std(),
    "Skewness": df[numeric_columns].skew(),
    "Kurtosis": df[numeric_columns].kurtosis()
})

print(distribution_summary)

distribution_summary.to_csv(
    os.path.join(EDA_FOLDER, "step_40_distribution_summary.csv")
)
```

```
======================================================================
STEP 40 — DISTRIBUTION ANALYSIS
======================================================================
                                      Mean    Median     Std_Dev   Skewness  \
is_canceled                       0.370416     0.000    0.482918   0.536678
lead_time                       104.011416    69.000  106.863097   1.346550
arrival_date_year              2024.000000  2024.000    0.000000   0.000000
arrival_date_week_number         26.375835    26.000   15.021596   0.003881
arrival_date_day_of_month        15.723394    16.000    8.805079   0.002373
stays_in_weekend_nights           0.927599     1.000    0.998613   1.380046
stays_in_week_nights              2.500302     2.000    1.908286   2.862249
adults                            1.856403     2.000    0.579261  18.317805
children                          0.103886     0.000    0.398555   4.112670
babies                            0.007949     0.000    0.097436  24.646545
is_repeated_guest                 0.031912     0.000    0.175767   5.326315
previous_cancellations            0.087118     0.000    0.844336  24.458049
previous_bookings_not_canceled    0.137097     0.000    1.497437  23.539800
booking_changes                   0.221124     0.000    0.652306   6.000270
agent                            74.828319     9.000  107.141953   1.300287
company                          10.775157     0.000   53.943884   5.918406
days_in_waiting_list              2.321149     0.000   17.594721  11.944353
adr                             101.831122    94.575   50.535790  10.530214
required_car_parking_spaces       0.062518     0.000    0.245291   4.163233
total_of_special_requests         0.571363     0.000    0.792798   1.349189
total_nights                      3.427900     3.000    2.557439   3.308771
total_guests                      1.968239     2.000    0.722394  10.154167
arrival_month_num                 6.484136     7.000    3.437030  -0.001028

                                   Kurtosis
is_canceled                       -1.712005
lead_time                          1.696449
arrival_date_year                  0.000000
arrival_date_week_number          -1.200734
arrival_date_day_of_month         -1.200683
stays_in_weekend_nights            7.174066
stays_in_week_nights              24.284555
adults                          1352.115116
children                          18.674464
babies                          1633.948235
is_repeated_guest                 26.370077
previous_cancellations           674.073693
previous_bookings_not_canceled   767.245210
booking_changes                   79.393605
agent                              0.512689
company                           37.753758
days_in_waiting_list             186.793070
adr                             1013.189851
required_car_parking_spaces       29.998056
total_of_special_requests          1.492565
total_nights                      28.873721
total_guests                     554.650474
arrival_month_num                 -1.206790
```

In [19]:
```
# ============================================================
# STEP 41 — DISTRIBUTION INTERPRETATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 41 — DISTRIBUTION INTERPRETATION")
print("=" * 70)
```

```
distribution_interpretation = []

for column in numeric_columns:

    skew_value = df[column].skew()

    if skew_value > 1:
        interpretation = "Highly positively/right skewed"

    elif skew_value > 0.5:
        interpretation = "Moderately positively/right skewed"

    elif skew_value < -1:
        interpretation = "Highly negatively/left skewed"

    elif skew_value < -0.5:
        interpretation = "Moderately negatively/left skewed"

    else:
        interpretation = "Approximately symmetric"

    distribution_interpretation.append({
        "Feature": column,
        "Skewness": skew_value,
        "Interpretation": interpretation
    })

    print(f"{column}: {interpretation}")

distribution_interpretation_df = pd.DataFrame(
    distribution_interpretation
)

distribution_interpretation_df.to_csv(
    os.path.join(
        EDA_FOLDER,
        "step_41_distribution_interpretation.csv"
    ),
    index=False
)
```

```
======================================================================
STEP 41 — DISTRIBUTION INTERPRETATION
======================================================================
is_canceled: Moderately positively/right skewed
lead_time: Highly positively/right skewed
arrival_date_year: Approximately symmetric
arrival_date_week_number: Approximately symmetric
arrival_date_day_of_month: Approximately symmetric
stays_in_weekend_nights: Highly positively/right skewed
stays_in_week_nights: Highly positively/right skewed
adults: Highly positively/right skewed
children: Highly positively/right skewed
babies: Highly positively/right skewed
is_repeated_guest: Highly positively/right skewed
previous_cancellations: Highly positively/right skewed
previous_bookings_not_canceled: Highly positively/right skewed
booking_changes: Highly positively/right skewed
agent: Highly positively/right skewed
company: Highly positively/right skewed
days_in_waiting_list: Highly positively/right skewed
adr: Highly positively/right skewed
required_car_parking_spaces: Highly positively/right skewed
total_of_special_requests: Highly positively/right skewed
total_nights: Highly positively/right skewed
total_guests: Highly positively/right skewed
arrival_month_num: Approximately symmetric
```

In [20]:

```python
# ============================================================
# STEP 42 — UNIQUE VALUE ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 42 — UNIQUE VALUE ANALYSIS")
print("=" * 70)

unique_values = df.nunique()

print(unique_values)

unique_values.to_csv(
    os.path.join(EDA_FOLDER, "step_42_unique_values.csv"),
    header=["Unique_Values"]
)
```

```
======================================================================
STEP 42 — UNIQUE VALUE ANALYSIS
======================================================================
hotel                                30
is_canceled                           2
lead_time                           479
arrival_date_year                     1
arrival_date_month                   12
arrival_date_week_number             52
arrival_date_day_of_month            31
stays_in_weekend_nights              17
stays_in_week_nights                 35
adults                               14
children                              5
babies                                5
meal                                  5
country                             178
market_segment                        8
distribution_channel                  5
is_repeated_guest                     2
previous_cancellations               15
previous_bookings_not_canceled       73
reserved_room_type                   10
assigned_room_type                   12
booking_changes                      21
deposit_type                          3
agent                               334
company                             353
days_in_waiting_list                128
customer_type                         4
adr                                8879
required_car_parking_spaces           5
total_of_special_requests             6
reservation_status                    3
reservation_status_date           75629
city                                 15
total_nights                         45
total_guests                         15
arrival_month_num                    12
arrival_date                        366
dtype: int64
```

In [21]:

```
# ============================================================
# STEP 43 — CATEGORICAL FREQUENCY ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 43 — CATEGORICAL FREQUENCY ANALYSIS")
print("=" * 70)

for column in categorical_columns:

    print("\nColumn:", column)

    frequency = df[column].value_counts(dropna=False)

    print(frequency.head(20))

    safe_name = column.replace(" ", "_").replace("/", "_")
```

```
    frequency.to_csv(
        os.path.join(
            EDA_FOLDER,
            f"step_43_{safe_name}_frequency.csv"
        ),
        header=["Frequency"]
    )
```

```
======================================================================
STEP 43 — CATEGORICAL FREQUENCY ANALYSIS
======================================================================

Column: hotel
hotel
City Hotel - Ahmedabad      5405
City Hotel - Bhopal         5367
City Hotel - Jaipur         5343
City Hotel - Pune           5343
City Hotel - Hyderabad      5339
City Hotel - Kolkata        5339
City Hotel - Goa            5297
City Hotel - Chandigarh     5293
City Hotel - Lucknow        5290
City Hotel - Mumbai         5265
City Hotel - Delhi          5252
City Hotel - Chennai        5251
City Hotel - Bangalore      5210
City Hotel - Indore         5170
City Hotel - Kochi          5166
Resort Hotel - Bhopal       2737
Resort Hotel - Delhi        2726
Resort Hotel - Kochi        2723
Resort Hotel - Jaipur       2695
Resort Hotel - Bangalore    2687
Name: count, dtype: int64

Column: arrival_date_month
arrival_date_month
October      10349
July         10298
March        10216
May          10212
January      10126
August       10033
June          9811
September     9759
April         9757
November      9696
December      9582
February      9551
Name: count, dtype: int64

Column: meal
meal
BB           92310
HB           14463
SC           10650
Undefined     1169
FB             798
Name: count, dtype: int64

Column: country
country
PRT    48590
GBR    12129
FRA    10415
ESP     8568
DEU     7287
```

```
ITA     3766
IRL     3375
BEL     2342
BRA     2224
NLD     2104
USA     2097
CHE     1730
CN      1279
AUT     1263
SWE     1024
CHN      999
POL      919
ISR      669
RUS      632
NOR      607
Name: count, dtype: int64

Column: market_segment
market_segment
Online TA        56477
Offline TA/TO    24219
Groups           19811
Direct           12606
Corporate         5295
Complementary      743
Aviation           237
Undefined            2
Name: count, dtype: int64

Column: distribution_channel
distribution_channel
TA/TO        97870
Direct       14645
Corporate     6677
GDS            193
Undefined        5
Name: count, dtype: int64

Column: reserved_room_type
reserved_room_type
A    85994
D    19201
E     6535
F     2897
G     2094
B     1118
C      932
H      601
P       12
L        6
Name: count, dtype: int64

Column: assigned_room_type
assigned_room_type
A    74053
D    25322
E     7806
F     3751
G     2553
C     2375
```

```
B     2163
H      712
I      363
K      279
P       12
L        1
Name: count, dtype: int64

Column: deposit_type
deposit_type
No Deposit    104641
Non Refund     14587
Refundable       162
Name: count, dtype: int64

Column: customer_type
customer_type
Transient          89613
Transient-Party    25124
Contract            4076
Group                577
Name: count, dtype: int64

Column: reservation_status
reservation_status
Check-Out    75166
Canceled     43017
No-Show       1207
Name: count, dtype: int64

Column: reservation_status_date
reservation_status_date
2024-01-13 19:56:56.304684686    8
2024-01-20 23:29:47.169672247    8
2024-08-03 07:50:56.271515800    7
2024-05-14 14:11:56.749449278    7
2024-02-20 02:48:47.968238279    7
2024-08-29 09:12:43.577884060    7
2024-04-18 09:40:51.210748058    7
2024-03-15 11:36:03.123905887    7
2024-03-11 03:11:24.877166238    7
2024-08-03 21:07:46.505289432    7
2024-10-23 04:39:14.025915284    6
2024-08-02 10:56:14.964192680    6
2024-12-10 12:50:13.785189588    6
2024-01-20 09:24:31.341580882    6
2024-06-24 20:13:44.397557564    6
2024-10-21 13:54:45.498664032    6
2024-02-07 13:58:53.722537252    6
2024-11-05 14:45:50.303629312    6
2024-08-31 04:21:20.801413864    6
2024-09-28 11:03:41.477690572    6
Name: count, dtype: int64

Column: city
city
Bhopal        8104
Jaipur        8038
Ahmedabad     8028
Pune          7992
```

```
Hyderabad     7981
Chandigarh    7978
Delhi         7978
Kolkata       7976
Goa           7973
Mumbai        7935
Chennai       7925
Bangalore     7897
Kochi         7889
Lucknow       7869
Indore        7827
Name: count, dtype: int64

Column: arrival_date
arrival_date
2024-03-22    383
2024-07-20    380
2024-04-07    373
2024-05-17    370
2024-07-02    366
2024-06-05    362
2024-07-30    362
2024-04-10    362
2024-07-14    361
2024-10-15    358
2024-01-11    357
2024-01-15    356
2024-10-29    356
2024-07-03    354
2024-12-19    354
2024-08-03    354
2024-10-25    353
2024-04-28    352
2024-07-15    352
2024-05-30    351
Name: count, dtype: int64
```

In [22]:

```
# ============================================================
# STEP 44 — CANCELLATION TARGET ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 44 — CANCELLATION TARGET ANALYSIS")
print("=" * 70)

target_column = "is_canceled"

if target_column in df.columns:

    target_distribution = df[target_column].value_counts()

    target_percentage = (
        df[target_column].value_counts(normalize=True) * 100
    )

    target_summary = pd.DataFrame({
        "Count": target_distribution,
        "Percentage": target_percentage
    })
```

```
    print(target_summary)

    target_summary.to_csv(
        os.path.join(EDA_FOLDER, "step_44_target_distribution.csv")
    )

else:
    raise ValueError("Target column is_canceled not found.")
```

```
======================================================================
STEP 44 — CANCELLATION TARGET ANALYSIS
======================================================================
             Count  Percentage
is_canceled
0            75166   62.958372
1            44224   37.041628
```

In [23]:

```
# ============================================================
# STEP 45 — CORRELATION MATRIX
# ============================================================

print("\n" + "=" * 70)
print("STEP 45 — CORRELATION MATRIX")
print("=" * 70)

correlation_matrix = df.select_dtypes(
    include=np.number
).corr()

print(correlation_matrix)

correlation_matrix.to_csv(
    os.path.join(EDA_FOLDER, "step_45_correlation_matrix.csv")
)
```

```
======================================================================
STEP 45 — CORRELATION MATRIX
======================================================================
                                is_canceled  lead_time  arrival_date_year  \
is_canceled                        1.000000   0.293123                NaN
lead_time                          0.293123   1.000000                NaN
arrival_date_year                       NaN        NaN                NaN
arrival_date_week_number           0.000500   0.004204                NaN
arrival_date_day_of_month         -0.003858  -0.000228                NaN
stays_in_weekend_nights           -0.001791   0.085671                NaN
stays_in_week_nights               0.024765   0.165799                NaN
adults                             0.060017   0.119519                NaN
children                           0.005036  -0.037613                NaN
babies                            -0.032491  -0.020915                NaN
is_repeated_guest                 -0.084793  -0.124410                NaN
previous_cancellations             0.110133   0.086042                NaN
previous_bookings_not_canceled    -0.057358  -0.073548                NaN
booking_changes                   -0.144381   0.000149                NaN
agent                             -0.046529  -0.012640                NaN
company                           -0.082995  -0.086250                NaN
days_in_waiting_list               0.054186   0.170084                NaN
adr                                0.047557  -0.063077                NaN
required_car_parking_spaces       -0.195498  -0.116451                NaN
total_of_special_requests         -0.234658  -0.095712                NaN
total_nights                       0.017779   0.157167                NaN
total_guests                       0.046522   0.072265                NaN
arrival_month_num                  0.000583   0.003841                NaN

                                arrival_date_week_number  \
is_canceled                                     0.000500
lead_time                                       0.004204
arrival_date_year                                    NaN
arrival_date_week_number                        1.000000
arrival_date_day_of_month                       0.071735
stays_in_weekend_nights                        -0.003025
stays_in_week_nights                           -0.002134
adults                                          0.001395
children                                        0.003525
babies                                          0.003893
is_repeated_guest                              -0.002788
previous_cancellations                         -0.000612
previous_bookings_not_canceled                 -0.001842
booking_changes                                 0.002116
agent                                          -0.001734
company                                        -0.002781
days_in_waiting_list                            0.007363
adr                                             0.000892
required_car_parking_spaces                    -0.004556
total_of_special_requests                      -0.001465
total_nights                                   -0.002774
total_guests                                    0.003589
arrival_month_num                               0.982385

                                arrival_date_day_of_month  \
is_canceled                                     -0.003858
lead_time                                       -0.000228
arrival_date_year                                     NaN
arrival_date_week_number                         0.071735
arrival_date_day_of_month                        1.000000
stays_in_weekend_nights                          0.001203
```

```
stays_in_week_nights                0.001929
adults                             -0.000779
children                           -0.000220
babies                             -0.003354
is_repeated_guest                  -0.001278
previous_cancellations             -0.002542
previous_bookings_not_canceled     -0.000117
booking_changes                     0.001535
agent                              -0.000939
company                            -0.002446
days_in_waiting_list               -0.004750
adr                                -0.001399
required_car_parking_spaces        -0.004174
total_of_special_requests           0.007877
total_nights                        0.001909
total_guests                       -0.001198
arrival_month_num                   0.002959

                                stays_in_weekend_nights  stays_in_week_nights  \
is_canceled                                   -0.001791              0.024765
lead_time                                      0.085671              0.165799
arrival_date_year                                   NaN                   NaN
arrival_date_week_number                      -0.003025             -0.002134
arrival_date_day_of_month                      0.001203              0.001929
stays_in_weekend_nights                        1.000000              0.498969
stays_in_week_nights                           0.498969              1.000000
adults                                         0.091871              0.092976
children                                       0.045794              0.044203
babies                                         0.018483              0.020191
is_repeated_guest                             -0.087239             -0.097245
previous_cancellations                        -0.012775             -0.013992
previous_bookings_not_canceled                -0.042715             -0.048743
booking_changes                                0.063281              0.096209
agent                                          0.161427              0.195135
company                                       -0.079977             -0.043641
days_in_waiting_list                          -0.054151             -0.002020
adr                                            0.049342              0.065237
required_car_parking_spaces                   -0.018554             -0.024859
total_of_special_requests                      0.072671              0.068192
total_nights                                   0.762790              0.941005
total_guests                                   0.101426              0.101665
arrival_month_num                             -0.003197             -0.002088

                                  adults  children    babies  ...  \
is_canceled                     0.060017  0.005036 -0.032491  ...
lead_time                       0.119519 -0.037613 -0.020915  ...
arrival_date_year                    NaN       NaN       NaN  ...
arrival_date_week_number        0.001395  0.003525  0.003893  ...
arrival_date_day_of_month      -0.000779 -0.000220 -0.003354  ...
stays_in_weekend_nights         0.091871  0.045794  0.018483  ...
stays_in_week_nights            0.092976  0.044203  0.020191  ...
adults                          1.000000  0.030440  0.018146  ...
children                        0.030440  1.000000  0.024030  ...
babies                          0.018146  0.024030  1.000000  ...
is_repeated_guest              -0.146426 -0.032858 -0.008943  ...
previous_cancellations         -0.006738 -0.024729 -0.007501  ...
previous_bookings_not_canceled -0.107983 -0.021072 -0.006550  ...
booking_changes                -0.051673  0.048952  0.083440  ...
agent                           0.024994  0.050581  0.030266  ...
company                        -0.166778 -0.042622 -0.009459  ...
```

```
days_in_waiting_list           -0.008283 -0.033271 -0.010621  ...
adr                             0.230641  0.324853  0.029186  ...
required_car_parking_spaces     0.014785  0.056255  0.037383  ...
total_of_special_requests       0.122884  0.081736  0.097889  ...
total_nights                    0.105249  0.050865  0.022283  ...
total_guests                    0.821105  0.579364  0.162688  ...
arrival_month_num               0.000892  0.003542  0.004066  ...

                                booking_changes     agent   company  \
is_canceled                          -0.144381 -0.046529 -0.082995
lead_time                             0.000149 -0.012640 -0.086250
arrival_date_year                          NaN       NaN       NaN
arrival_date_week_number              0.002116 -0.001734 -0.002781
arrival_date_day_of_month             0.001535 -0.000939 -0.002446
stays_in_weekend_nights               0.063281  0.161427 -0.079977
stays_in_week_nights                  0.096209  0.195135 -0.043641
adults                               -0.051673  0.024994 -0.166778
children                              0.048952  0.050581 -0.042622
babies                                0.083440  0.030266 -0.009459
is_repeated_guest                     0.012092 -0.052264  0.159723
previous_cancellations               -0.026993 -0.018192 -0.001190
previous_bookings_not_canceled        0.011608 -0.046296  0.110817
booking_changes                       1.000000  0.036478  0.088863
agent                                 0.036478  1.000000 -0.121536
company                               0.088863 -0.121536  1.000000
days_in_waiting_list                 -0.011634 -0.040853 -0.022986
adr                                   0.019618  0.016707 -0.128470
required_car_parking_spaces           0.065620  0.119158  0.038299
total_of_special_requests             0.052833  0.060696 -0.091066
total_nights                          0.096498  0.208637 -0.063793
total_guests                         -0.003173  0.052030 -0.158524
arrival_month_num                     0.002144 -0.001534 -0.002681

                                days_in_waiting_list       adr  \
is_canceled                                 0.054186  0.047557
lead_time                                   0.170084 -0.063077
arrival_date_year                                NaN       NaN
arrival_date_week_number                    0.007363  0.000892
arrival_date_day_of_month                  -0.004750 -0.001399
stays_in_weekend_nights                    -0.054151  0.049342
stays_in_week_nights                       -0.002020  0.065237
adults                                     -0.008283  0.230641
children                                   -0.033271  0.324853
babies                                     -0.010621  0.029186
is_repeated_guest                          -0.022235 -0.134314
previous_cancellations                      0.005929 -0.065646
previous_bookings_not_canceled             -0.009397 -0.072144
booking_changes                            -0.011634  0.019618
agent                                      -0.040853  0.016707
company                                    -0.022986 -0.128470
days_in_waiting_list                        1.000000 -0.040756
adr                                        -0.040756  1.000000
required_car_parking_spaces                -0.030600  0.056628
total_of_special_requests                  -0.082730  0.172185
total_nights                               -0.022652  0.067945
total_guests                               -0.026431  0.368105
arrival_month_num                           0.007533  0.001381

                                required_car_parking_spaces  \
is_canceled                                       -0.195498
```

```
lead_time                                 -0.116451
arrival_date_year                               NaN
arrival_date_week_number                  -0.004556
arrival_date_day_of_month                 -0.004174
stays_in_weekend_nights                   -0.018554
stays_in_week_nights                      -0.024859
adults                                     0.014785
children                                   0.056255
babies                                     0.037383
is_repeated_guest                          0.077090
previous_cancellations                    -0.018492
previous_bookings_not_canceled             0.047653
booking_changes                            0.065620
agent                                      0.119158
company                                    0.038299
days_in_waiting_list                      -0.030600
adr                                        0.056628
required_car_parking_spaces                1.000000
total_of_special_requests                  0.082626
total_nights                              -0.025794
total_guests                               0.047934
arrival_month_num                         -0.004049

                              total_of_special_requests  total_nights  \
is_canceled                                   -0.234658      0.017779
lead_time                                     -0.095712      0.157167
arrival_date_year                                   NaN           NaN
arrival_date_week_number                      -0.001465     -0.002774
arrival_date_day_of_month                      0.007877      0.001909
stays_in_weekend_nights                        0.072671      0.762790
stays_in_week_nights                           0.068192      0.941005
adults                                         0.122884      0.105249
children                                       0.081736      0.050865
babies                                         0.097889      0.022283
is_repeated_guest                              0.013050     -0.106626
previous_cancellations                        -0.048384     -0.015429
previous_bookings_not_canceled                 0.037824     -0.053049
booking_changes                                0.052833      0.096498
agent                                          0.060696      0.208637
company                                       -0.091066     -0.063793
days_in_waiting_list                          -0.082730     -0.022652
adr                                            0.172185      0.067945
required_car_parking_spaces                    0.082626     -0.025794
total_of_special_requests                      1.000000      0.079259
total_nights                                   0.079259      1.000000
total_guests                                   0.156834      0.115463
arrival_month_num                             -0.002414     -0.002806

                              total_guests  arrival_month_num
is_canceled                       0.046522           0.000583
lead_time                         0.072265           0.003841
arrival_date_year                      NaN                NaN
arrival_date_week_number          0.003589           0.982385
arrival_date_day_of_month        -0.001198           0.002959
stays_in_weekend_nights           0.101426          -0.003197
stays_in_week_nights              0.101665          -0.002088
adults                            0.821105           0.000892
children                          0.579364           0.003542
babies                            0.162688           0.004066
is_repeated_guest                -0.136748          -0.002670
```

```
previous_cancellations             -0.020058          -0.000763
previous_bookings_not_canceled     -0.099097          -0.001379
booking_changes                    -0.003173           0.002144
agent                               0.052030          -0.001534
company                            -0.158524          -0.002681
days_in_waiting_list               -0.026431           0.007533
adr                                 0.368105           0.001381
required_car_parking_spaces         0.047934          -0.004049
total_of_special_requests           0.156834          -0.002414
total_nights                        0.115463          -0.002806
total_guests                        1.000000           0.003218
arrival_month_num                   0.003218           1.000000

[23 rows x 23 columns]
```

In [24]:

```
# ============================================================
# STEP 46 — STRONG CORRELATION ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 46 — STRONG CORRELATIONS")
print("=" * 70)

correlation_pairs = []

for i in range(len(correlation_matrix.columns)):

    for j in range(i + 1, len(correlation_matrix.columns)):

        feature_1 = correlation_matrix.columns[i]
        feature_2 = correlation_matrix.columns[j]

        correlation_value = correlation_matrix.iloc[i, j]

        correlation_pairs.append({
            "Feature_1": feature_1,
            "Feature_2": feature_2,
            "Correlation": correlation_value,
            "Absolute_Correlation": abs(correlation_value)
        })

correlation_pairs_df = pd.DataFrame(correlation_pairs)

correlation_pairs_df = correlation_pairs_df.sort_values(
    "Absolute_Correlation",
    ascending=False
)

print(correlation_pairs_df.head(20))

correlation_pairs_df.to_csv(
    os.path.join(EDA_FOLDER, "step_46_strong_correlations.csv"),
    index=False
)
```

```
======================================================================
STEP 46 — STRONG CORRELATIONS
======================================================================
                    Feature_1                       Feature_2  Correlation  \
81   arrival_date_week_number              arrival_month_num     0.982385
130      stays_in_week_nights                   total_nights     0.941005
146                    adults                   total_guests     0.821105
114   stays_in_weekend_nights                   total_nights     0.762790
160                  children                   total_guests     0.579364
100   stays_in_weekend_nights           stays_in_week_nights     0.498969
176         is_repeated_guest  previous_bookings_not_canceled     0.418056
241                       adr                   total_guests     0.368105
156                  children                            adr     0.324853
0                 is_canceled                      lead_time     0.293123
18                is_canceled      total_of_special_requests    -0.234658
142                    adults                            adr     0.230641
222                    agent                   total_nights     0.208637
17                is_canceled    required_car_parking_spaces    -0.195498
124      stays_in_week_nights                           agent     0.195135
239                       adr      total_of_special_requests     0.172185
36                  lead_time           days_in_waiting_list     0.170084
140                    adults                        company    -0.166778
26                  lead_time           stays_in_week_nights     0.165799
173                    babies                   total_guests     0.162688

     Absolute_Correlation
81               0.982385
130              0.941005
146              0.821105
114              0.762790
160              0.579364
100              0.498969
176              0.418056
241              0.368105
156              0.324853
0                0.293123
18               0.234658
142              0.230641
222              0.208637
17               0.195498
124              0.195135
239              0.172185
36               0.170084
140              0.166778
26               0.165799
173              0.162688
```

In [25]:

```
# ============================================================
# STEP 47 — FEATURES CORRELATED WITH CANCELLATION
# ============================================================

print("\n" + "=" * 70)
print("STEP 47 — FEATURES CORRELATED WITH CANCELLATION")
print("=" * 70)

if target_column in correlation_matrix.columns:

    cancellation_correlations = (
        correlation_matrix[target_column]
        .drop(target_column)
```

```
        .sort_values(key=abs, ascending=False)
    )

    print(cancellation_correlations)

    cancellation_correlations.to_csv(
        os.path.join(
            EDA_FOLDER,
            "step_47_cancellation_correlations.csv"
        ),
        header=["Correlation"]
    )
```

```
======================================================================
STEP 47 — FEATURES CORRELATED WITH CANCELLATION
======================================================================
lead_time                          0.293123
total_of_special_requests         -0.234658
required_car_parking_spaces       -0.195498
booking_changes                   -0.144381
previous_cancellations             0.110133
is_repeated_guest                 -0.084793
company                           -0.082995
adults                             0.060017
previous_bookings_not_canceled    -0.057358
days_in_waiting_list               0.054186
adr                                0.047557
agent                             -0.046529
total_guests                       0.046522
babies                            -0.032491
stays_in_week_nights               0.024765
total_nights                       0.017779
children                           0.005036
arrival_date_day_of_month         -0.003858
stays_in_weekend_nights           -0.001791
arrival_month_num                  0.000583
arrival_date_week_number           0.000500
arrival_date_year                       NaN
Name: is_canceled, dtype: float64
```

In [26]:

```
# ============================================================
# STEP 48 — ANOMALY / OUTLIER DETECTION
# ============================================================

print("\n" + "=" * 70)
print("STEP 48 — ANOMALY DETECTION")
print("=" * 70)

anomaly_summary = []

for column in numeric_columns:

    values = df[column].dropna()

    mean_value = values.mean()
    std_value = values.std()

    if pd.notna(std_value) and std_value != 0:

        z_scores = (values - mean_value) / std_value
```

```
        anomaly_count = (abs(z_scores) > 3).sum()

    else:
        anomaly_count = 0

    anomaly_summary.append({
        "Feature": column,
        "Anomalies_ZScore_GT_3": int(anomaly_count)
    })

    print(f"{column}: {anomaly_count} anomalies")

anomaly_df = pd.DataFrame(anomaly_summary)

anomaly_df.to_csv(
    os.path.join(EDA_FOLDER, "step_48_anomaly_analysis.csv"),
    index=False
)
```

```
======================================================================
STEP 48 — ANOMALY DETECTION
======================================================================
is_canceled: 0 anomalies
lead_time: 1454 anomalies
arrival_date_year: 0 anomalies
arrival_date_week_number: 0 anomalies
arrival_date_day_of_month: 0 anomalies
stays_in_weekend_nights: 2199 anomalies
stays_in_week_nights: 1669 anomalies
adults: 481 anomalies
children: 3729 anomalies
babies: 917 anomalies
is_repeated_guest: 3810 anomalies
previous_cancellations: 317 anomalies
previous_bookings_not_canceled: 936 anomalies
booking_changes: 1570 anomalies
agent: 971 anomalies
company: 3575 anomalies
days_in_waiting_list: 1871 anomalies
adr: 1138 anomalies
required_car_parking_spaces: 7416 anomalies
total_of_special_requests: 2877 anomalies
total_nights: 1720 anomalies
total_guests: 154 anomalies
arrival_month_num: 0 anomalies
```

In [27]:

```
# ============================================================
# STEP 49 — FEATURE VARIABILITY ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 49 — FEATURE VARIABILITY")
print("=" * 70)

variability = pd.DataFrame({
    "Mean": df[numeric_columns].mean(),
    "Std_Dev": df[numeric_columns].std(),
    "Variance": df[numeric_columns].var()
})
```

```
variability["Coefficient_of_Variation"] = (
    variability["Std_Dev"]
    / variability["Mean"].replace(0, np.nan)
)

variability = variability.sort_values(
    "Coefficient_of_Variation",
    ascending=False
)

print(variability)

variability.to_csv(
    os.path.join(EDA_FOLDER, "step_49_feature_variability.csv")
)
```

```
======================================================================
STEP 49 — FEATURE VARIABILITY
======================================================================
                                      Mean     Std_Dev      Variance  \
babies                            0.007949    0.097436      0.009494
previous_bookings_not_canceled    0.137097    1.497437      2.242317
previous_cancellations            0.087118    0.844336      0.712904
days_in_waiting_list              2.321149   17.594721    309.574203
is_repeated_guest                 0.031912    0.175767      0.030894
company                          10.775157   53.943884   2909.942594
required_car_parking_spaces       0.062518    0.245291      0.060168
children                          0.103886    0.398555      0.158846
booking_changes                   0.221124    0.652306      0.425503
agent                            74.828319  107.141953  11479.398119
total_of_special_requests         0.571363    0.792798      0.628529
is_canceled                       0.370416    0.482918      0.233210
stays_in_weekend_nights           0.927599    0.998613      0.997229
lead_time                       104.011416  106.863097  11419.721511
stays_in_week_nights              2.500302    1.908286      3.641554
total_nights                      3.427900    2.557439      6.540493
arrival_date_week_number         26.375835   15.021596    225.648350
arrival_date_day_of_month        15.723394    8.805079     77.529415
arrival_month_num                 6.484136    3.437030     11.813178
adr                             101.831122   50.535790   2553.866100
total_guests                      1.968239    0.722394      0.521853
adults                            1.856403    0.579261      0.335543
arrival_date_year              2024.000000    0.000000      0.000000

                                Coefficient_of_Variation
babies                                         12.258068
previous_bookings_not_canceled                 10.922470
previous_cancellations                          9.691887
days_in_waiting_list                            7.580177
is_repeated_guest                               5.507832
company                                         5.006320
required_car_parking_spaces                     3.923541
children                                        3.836451
booking_changes                                 2.949953
agent                                           1.431837
total_of_special_requests                       1.387557
is_canceled                                     1.303718
stays_in_weekend_nights                         1.076558
lead_time                                       1.027417
stays_in_week_nights                            0.763222
total_nights                                    0.746066
arrival_date_week_number                        0.569521
arrival_date_day_of_month                       0.559999
arrival_month_num                               0.530068
adr                                             0.496271
total_guests                                    0.367026
adults                                          0.312034
arrival_date_year                               0.000000
```

In [28]:

```
# ============================================================
# STEP 50 — MEAN COMPARISON BY CANCELLATION STATUS
# ============================================================

print("\n" + "=" * 70)
print("STEP 50 — MEAN BY CANCELLATION STATUS")
print("=" * 70)
```

```
grouped_means = df.groupby(target_column)[numeric_columns].mean().T

print(grouped_means)

grouped_means.to_csv(
    os.path.join(EDA_FOLDER, "step_50_means_by_cancellation.csv")
)
```

```
======================================================================
STEP 50 — MEAN BY CANCELLATION STATUS
======================================================================
is_canceled                                0            1
is_canceled                         0.000000     1.000000
lead_time                          79.984687   144.848815
arrival_date_year                2024.000000  2024.000000
arrival_date_week_number           26.370074    26.385628
arrival_date_day_of_month          15.749448    15.679111
stays_in_weekend_nights             0.928971     0.925267
stays_in_week_nights                2.464053     2.561912
adults                              1.829737     1.901728
children                            0.102347     0.106503
babies                              0.010377     0.003821
is_repeated_guest                   0.043344     0.012482
previous_cancellations              0.015792     0.208348
previous_bookings_not_canceled      0.202977     0.025122
booking_changes                     0.293364     0.098340
agent                              78.652196    68.329007
company                            14.209230     4.938382
days_in_waiting_list                1.589868     3.564083
adr                                99.987693   104.964333
required_car_parking_spaces         0.099300     0.000000
total_of_special_requests           0.714060     0.328826
total_nights                        3.393023     3.487179
total_guests                        1.942461     2.012052
arrival_month_num                   6.482599     6.486749
```

In [29]:
```
# ============================================================
# STEP 51 — MEDIAN COMPARISON BY CANCELLATION STATUS
# ============================================================

print("\n" + "=" * 70)
print("STEP 51 — MEDIAN BY CANCELLATION STATUS")
print("=" * 70)

grouped_medians = df.groupby(target_column)[numeric_columns].median().T

print(grouped_medians)

grouped_medians.to_csv(
    os.path.join(EDA_FOLDER, "step_51_medians_by_cancellation.csv")
)
```

```
======================================================================
STEP 51 — MEDIAN BY CANCELLATION STATUS
======================================================================
is_canceled                          0       1
is_canceled                        0.0     1.0
lead_time                         45.0   113.0
arrival_date_year               2024.0  2024.0
arrival_date_week_number          26.0    26.0
arrival_date_day_of_month         16.0    16.0
stays_in_weekend_nights            1.0     1.0
stays_in_week_nights               2.0     2.0
adults                             2.0     2.0
children                           0.0     0.0
babies                             0.0     0.0
is_repeated_guest                  0.0     0.0
previous_cancellations             0.0     0.0
previous_bookings_not_canceled     0.0     0.0
booking_changes                    0.0     0.0
agent                              9.0     9.0
company                            0.0     0.0
days_in_waiting_list               0.0     0.0
adr                               92.5    96.2
required_car_parking_spaces        0.0     0.0
total_of_special_requests          1.0     0.0
total_nights                       3.0     3.0
total_guests                       2.0     2.0
arrival_month_num                  7.0     7.0
```

In [30]:

```
# ============================================================
# STEP 52 — HOTEL TYPE CANCELLATION ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 52 — HOTEL TYPE ANALYSIS")
print("=" * 70)

if "hotel" in df.columns:

    hotel_analysis = df.groupby("hotel")[target_column].agg(
        Bookings="count",
        Cancellation_Rate="mean"
    )

    hotel_analysis["Cancellation_Percentage"] = (
        hotel_analysis["Cancellation_Rate"] * 100
    )

    print(hotel_analysis)

    hotel_analysis.to_csv(
        os.path.join(EDA_FOLDER, "step_52_hotel_analysis.csv")
    )
```

```
======================================================================
STEP 52 — HOTEL TYPE ANALYSIS
======================================================================
                           Bookings  Cancellation_Rate  \
hotel
City Hotel - Ahmedabad         5405           0.422757
City Hotel - Bangalore         5210           0.432438
City Hotel - Bhopal            5367           0.406372
City Hotel - Chandigarh        5293           0.416021
City Hotel - Chennai           5251           0.413635
City Hotel - Delhi             5252           0.412795
City Hotel - Goa               5297           0.414952
City Hotel - Hyderabad         5339           0.422926
City Hotel - Indore            5170           0.417988
City Hotel - Jaipur            5343           0.412128
City Hotel - Kochi             5166           0.409214
City Hotel - Kolkata           5339           0.425735
City Hotel - Lucknow           5290           0.419282
City Hotel - Mumbai            5265           0.415954
City Hotel - Pune              5343           0.416807
Resort Hotel - Ahmedabad       2623           0.274114
Resort Hotel - Bangalore       2687           0.282471
Resort Hotel - Bhopal          2737           0.270734
Resort Hotel - Chandigarh      2685           0.268156
Resort Hotel - Chennai         2674           0.278235
Resort Hotel - Delhi           2726           0.262656
Resort Hotel - Goa             2676           0.268311
Resort Hotel - Hyderabad       2642           0.299016
Resort Hotel - Indore          2657           0.283026
Resort Hotel - Jaipur          2695           0.278293
Resort Hotel - Kochi           2723           0.286449
Resort Hotel - Kolkata         2637           0.278347
Resort Hotel - Lucknow         2579           0.275688
Resort Hotel - Mumbai          2670           0.277903
Resort Hotel - Pune            2649           0.281616

                           Cancellation_Percentage
hotel
City Hotel - Ahmedabad                   42.275671
City Hotel - Bangalore                   43.243762
City Hotel - Bhopal                      40.637228
City Hotel - Chandigarh                  41.602116
City Hotel - Chennai                     41.363550
City Hotel - Delhi                       41.279513
City Hotel - Goa                         41.495186
City Hotel - Hyderabad                   42.292564
City Hotel - Indore                      41.798839
City Hotel - Jaipur                      41.212802
City Hotel - Kochi                       40.921409
City Hotel - Kolkata                     42.573516
City Hotel - Lucknow                     41.928166
City Hotel - Mumbai                      41.595442
City Hotel - Pune                        41.680704
Resort Hotel - Ahmedabad                 27.411361
Resort Hotel - Bangalore                 28.247116
Resort Hotel - Bhopal                    27.073438
Resort Hotel - Chandigarh                26.815642
Resort Hotel - Chennai                   27.823485
Resort Hotel - Delhi                     26.265591
Resort Hotel - Goa                       26.831091
```

```
Resort Hotel - Hyderabad                29.901590
Resort Hotel - Indore                   28.302597
Resort Hotel - Jaipur                   27.829314
Resort Hotel - Kochi                    28.644877
Resort Hotel - Kolkata                  27.834661
Resort Hotel - Lucknow                  27.568825
Resort Hotel - Mumbai                   27.790262
Resort Hotel - Pune                     28.161570
```

In [31]:
```
# ============================================================
# STEP 53 — CUSTOMER TYPE CANCELLATION ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 53 — CUSTOMER TYPE ANALYSIS")
print("=" * 70)

if "customer_type" in df.columns:

    customer_analysis = df.groupby("customer_type")[target_column].agg(
        Bookings="count",
        Cancellation_Rate="mean"
    )

    customer_analysis["Cancellation_Percentage"] = (
        customer_analysis["Cancellation_Rate"] * 100
    )

    print(customer_analysis)

    customer_analysis.to_csv(
        os.path.join(EDA_FOLDER, "step_53_customer_type_analysis.csv")
    )
```

```
======================================================================
STEP 53 — CUSTOMER TYPE ANALYSIS
======================================================================
                 Bookings  Cancellation_Rate  Cancellation_Percentage
customer_type
Contract             4076           0.309617                30.961727
Group                 577           0.102253                10.225303
Transient           89613           0.407463                40.746320
Transient-Party     25124           0.254299                25.429868
```

In [32]:
```
# ============================================================
# STEP 54 — DEPOSIT TYPE CANCELLATION ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 54 — DEPOSIT TYPE ANALYSIS")
print("=" * 70)

if "deposit_type" in df.columns:

    deposit_analysis = df.groupby("deposit_type")[target_column].agg(
        Bookings="count",
        Cancellation_Rate="mean"
    )

    deposit_analysis["Cancellation_Percentage"] = (
```

```
        deposit_analysis["Cancellation_Rate"] * 100
    )

    print(deposit_analysis)

    deposit_analysis.to_csv(
        os.path.join(EDA_FOLDER, "step_54_deposit_analysis.csv")
    )
```

```
======================================================================
STEP 54 — DEPOSIT TYPE ANALYSIS
======================================================================
              Bookings  Cancellation_Rate  Cancellation_Percentage
deposit_type
No Deposit      104641           0.283770                28.377022
Non Refund       14587           0.993624                99.362446
Refundable         162           0.222222                22.222222
```

In [33]:
```
# ============================================================
# STEP 55 — LEAD TIME AND ADR ANALYSIS
# ============================================================

print("\n" + "=" * 70)
print("STEP 55 — LEAD TIME AND ADR ANALYSIS")
print("=" * 70)

lead_adr_results = {}

for column in ["lead_time", "adr", "total_nights"]:

    if column in df.columns:

        lead_adr_results[column] = df.groupby(target_column)[column].agg(
            ["count", "mean", "median", "std"]
        )

        print(f"\n{column} by cancellation status:")
        print(lead_adr_results[column])

        lead_adr_results[column].to_csv(
            os.path.join(
                EDA_FOLDER,
                f"step_55_{column}_analysis.csv"
            )
        )
```

```
======================================================================
STEP 55 — LEAD TIME AND ADR ANALYSIS
======================================================================

lead_time by cancellation status:
             count        mean  median         std
is_canceled
0            75166   79.984687    45.0   91.109888
1            44224  144.848815   113.0  118.624829

adr by cancellation status:
             count        mean  median        std
is_canceled
0            75166   99.987693    92.5  49.206263
1            44224  104.964333    96.2  52.571142

total_nights by cancellation status:
             count      mean  median       std
is_canceled
0            75166  3.393023     3.0  2.577671
1            44224  3.487179     3.0  2.521601
```

In [34]:

```python
# ============================================================
# STEP 56 — FINAL EDA SUMMARY
# ============================================================

print("\n" + "=" * 70)
print("STEP 56 — FINAL EDA SUMMARY")
print("=" * 70)

eda_summary = {
    "Total_Records": len(df),
    "Total_Features": df.shape[1],
    "Numerical_Features": len(numeric_columns),
    "Categorical_Features": len(categorical_columns),
    "Missing_Values": int(df.isnull().sum().sum()),
    "Duplicate_Records": int(df.duplicated().sum()),
    "Cancelled_Bookings": int((df[target_column] == 1).sum()),
    "Not_Cancelled_Bookings": int((df[target_column] == 0).sum()),
    "Cancellation_Rate_Percentage": round(
        df[target_column].mean() * 100, 2
    )
}

for key, value in eda_summary.items():
    print(f"{key}: {value}")

pd.DataFrame(
    list(eda_summary.items()),
    columns=["Metric", "Value"]
).to_csv(
    os.path.join(EDA_FOLDER, "step_56_final_eda_summary.csv"),
    index=False
)

print("\nMEMBER 3 — STEPS 22–56 COMPLETED")
```

```
======================================================================
STEP 56 — FINAL EDA SUMMARY
======================================================================
Total_Records: 119390
Total_Features: 37
Numerical_Features: 23
Categorical_Features: 14
Missing_Values: 0
Duplicate_Records: 0
Cancelled_Bookings: 44224
Not_Cancelled_Bookings: 75166
Cancellation_Rate_Percentage: 37.04

MEMBER 3 — STEPS 22–56 COMPLETED
```